

Comparison and Final Conclusion

1. Comparison Table

Feature	Adjacency Matrix	Adjacency List
Storage method	2D array	Linked lists
Space complexity	$O(V^2)$	$O(V + E)$
BFS time	$O(V^2)$	$O(V + E)$
DFS time	$O(V^2)$	$O(V + E)$
BFS operations	36	12
DFS operations	36	12
Edge checking	$O(1)$	$O(\text{degree})$
E-F edge check	1 operation	2 operations
Suitable for	Dense graphs	Sparse graphs

2. Final Conclusion

The given social network contains 6 vertices and 6 edges, so it is a relatively sparse graph. The adjacency matrix requires a 6×6 structure containing 36 entries, while the adjacency list stores the vertices and their actual connections. From the execution results, BFS required 36 neighbor checks using the adjacency matrix compared with 12 using the adjacency list. Similarly, DFS required 36 checks with the matrix and 12 with the list. However, for checking the specific E-F edge, the matrix required only 1 operation, while the adjacency list required 2 operations. Therefore, for this sparse social network, the adjacency list is more suitable overall because it requires less storage and avoids checking non-existent connections during traversal. The adjacency matrix is advantageous when frequent direct edge checking is required.